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Keywords

Analog circuits, Computer-aided design, Multi-stage amplifier, computing and processing, electronic design automation, large language model, ring oscillator

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LADAC: Large Language Model-driven Auto-Designer for Analog Circuits

Chengjie Liu, Yijiang Liu, Yuan Du, *Senior Member, IEEE,* and Li Du, *Member, IEEE,*

***Abstract*—Large Language Models (LLMs) have shown their capabilities in solving tasks across various domains. Recently, they have found applications in automating the design of digital circuits. However, there remains a dearth of research investigating the use of LLMs in automating the design of analog circuits. In this work, we propose LADAC: Large Language Model-driven Auto-Designer for Analog Circuits, to assist analog circuit design. LADAC designates an LLM as a decision-making agent, which formulates optimized design strategies based on user specifications and foundational principles of analog circuit design expertise. We applied LADAC to design two kinds of amplifiers with open-loop gain over 80dB, and a ring oscillator with 100MHz oscillation frequency successfully, showing that LADAC is capable of designing analog circuits. Furthermore, it demonstrates the LLMs' potential for designing a broader range of analog circuits.**

***Index Terms*—Analog circuits, Computer-aided design, Electronic design automation, Large language model, Multi-stage amplifier, Ring oscillator**

I. Introduction

THE advancements of large language models (LLMs) have enabled them to exhibit remarkable performance across a diverse range of tasks through their common sense reasoning skills. These tasks include natural language processing [1]–[4], content creation [5], information retrieval [6], and code generation [7]. This has unveiled the potential for LLMs to understand and achieve automated circuit design process.

For digital circuits, several researches about exploring utilizing LLMs for achieving their auto-design. BlockLove et al. [8] chatted with an LLM and let it design an 8-bit accumulator-based microprocessor. Shailja et al. [9] developed a training corpus specifically for Verilog code, which was subsequently utilized to train five distinct fine-tuning LLMs. He et al. [10] developed ChatEDA, an interfacing framework and methodology for electronic design automation (EDA). NVIDIA [11] pretrained and fine-tuned a domain-adapted LLM, ChipNeMo, which demonstrated a better understanding of digital circuits and design capabilities than the general-purpose LLMs of comparable magnitude to a 13B model. These works demonstrate that LLMs are equipped with some basic knowledge of digital circuits and are capable of writing code for digital circuits.

While digital circuits have experienced success in LLM-driven auto design, using LLMs to assist analog circuit design has not yet developed due to these distinct challenges:

- Analog circuits, in comparison to digital circuits, possess intricate interconnections that cannot be effectively conveyed through codes, making it challenging for LLMs to comprehend them thoroughly;
- In the process of analog circuit design, iterative transistor sizing to achieve the expected performance metrics cannot be readily generated by LLMs in a single attempt;
- Current LLMs, despite their advancements, lack a comprehensive understanding of analog circuits, making it difficult for them to capture the inherent complexity of analog circuit structures.

To address these challenges, we propose LADAC: a Large Language Model-driven Auto-Designer for Analog Circuits. LADAC designates an LLM as a decision-making agent to facilitate its systematic execution of tasks such as analog knowledge retrieval, transistor sizing, and simulation, which entail the application of reasoning. LADAC formulates optimized design strategies informed by user specifications and grounded in foundational principles of analog expertise. Subsequently, it utilizes measurement description language (MDL) to extract simulation outcomes, obtains the performance metrics of the circuit with the current transistor parameters.

Our primary contributions are:

- The development of a domain-specified LLM agent capable of analysis and decision-making in the analog circuit design process, while incorporating appropriate external tools;
- The construction of a local knowledge library rich in domain-specific insights into analog circuit design, bridging the knowledge gap in existing LLMs;
- The development of interactive tools that allow LADAC to autonomously size transistors in the circuit and obtain simulation outcomes, enhancing its design capabilities;
- Demonstrating the practicality and potential of LADAC through the successful design of two kinds of amplifiers with open-loop gain over 80dB, and a ring oscillator [12] with 100MHz oscillation frequency, setting a precedent for further LLM advancements in the field.

Overall, our work demonstrates the potential and effectiveness of utilizing a domain-specified LLM agent equipped with interactive tools, supplemented by a local knowledge library, in automating the design of analog circuits. This paper is organized as follows: Section II illustrates the background of LLMs and related applications. Section III outlines the overall workflow of LADAC. Section IV exhibits the experiment of applying LADAC to the design of a 2-stage amplifier, a 3-stage NMCNR amplifier, and a ring oscillator. Finally, Section V concludes by summarizing our work, pointing out the challenges and opportunities of LLMs-assisted analog circuit design, and looking forward to prospects.

II. Background of the LLMs and Agents

In this section, we will provide an introduction to the background of utilizing LLMs to assist digital circuit design. Subsequently, we will present a list of reasons why the conversational approach of LLMs is not suitable for analog circuit design. Finally, we will introduce other research on agents that can be leveraged for the purpose of automated analog circuit design.

A. LLMs on Hardware Description Language (HDL)

The automated generation of hardware circuits using an LLM began by Pearce et al. [13]. They utilized a GPT-2 model and demonstrated that LLMs can generate syntactically correct Verilog code, although there was still room for improvement. Since then, significant progress has been made with the development of larger language models. [9], [10], [14], [15] have been actively investigating the generation of Verilog code at various levels. A milestone was achieved by Blocklove et al [8], who achieved the entire Verilog code generation process using LLMs, resulting in the successful tape out of a microprocessor using a Skywater 130nm shuttle. This breakthrough showcases the potential of LLMs in automating the generation of Verilog code for hardware circuits.

In contrast to digital circuits, which are composed of code and can be understood by LLMs, analog circuits pose a greater challenge for LLMs due to their complex connections and parameters. As a result, LLMs struggle to fully comprehend the intricacies of analog circuits, making it difficult for them to achieve the automated design of analog circuits.

B. Lack of analog knowledge

In this part, we aim to highlight the limitations of the current LLM knowledge library in understanding analog circuits. To assess the understanding of LLMs for analog circuits, we conducted a series of question-and-answer experiments using one of the state-of-the-art LLMs, GPT4 [1].

We conducted experiments to assess the understanding of the GPT4 in the context of analog circuits. Firstly, we tested its comprehension of RC filters, a common passive structure in analog circuits. We requested the GPT4 to generate a SPICE netlist file for a 3rd-order low-pass RC filter with a cut-off frequency of 10MHz. However, GPT4 generated an incorrect passive network that did not match our expectations, despite the correct values of resistors and capacitors as shown in Fig. 1.

Next, we try to communicate with the GPT4 and make it generate a circuit made of active devices: a five-transistor operational transconductance amplifier (5-T OTA). The circuit netlist generated by GPT-4 still exhibits multiple deviations from the expected outcomes. Its reply and the corresponding schematic are shown in Fig. 2.

Analyzing the generated 3rd-order RC low-pass filter and the 5-T OTA netlist file from GPT4, it is evident that the LLM failed to accurately understand and design the expected circuits. This highlights the current LLM's limited comprehensibility for analog circuits, particularly when it involves

GPT-4's response to a 3rd-order RC-filter netlist

Input: Please generate a SPICE netlist for a 3rd-order low pass RC-filter, whose cut-off frequency is 10MHz

Output:

```
*3rd Order Low Pass RC Filter

.param fc=10Meg R=1
.param C={1/(2*pi*R*fc)}

Vinput in 0 AC 1 SIN(0V {sqrt(2)}V {fc})

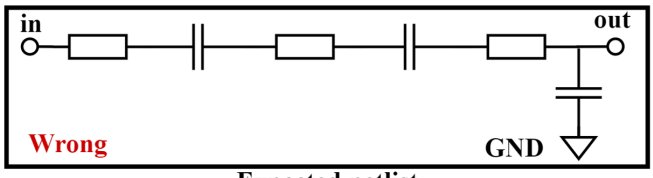
R1 in n1 R
C1 n1 n2 C

R2 n2 n3 R
C2 n3 n4 C

R3 n4 out R
C3 out GND C

.tran .5us {100/fc}
.ac dec 20 .01Meg 100Meg
.probe ac V(out)
.end
```

Netlist generated by GPT-4



Expected netlist

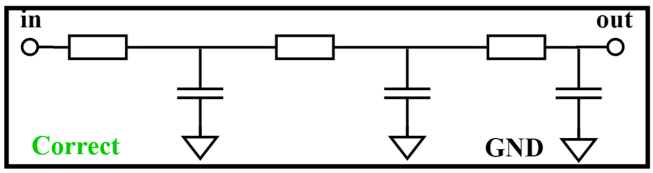


Fig. 1. SPICE netlist file for a 3rd-order low-pass RC filter generated by GPT4 and comparison between the generated wrong circuit with the expected one.

circuits with active devices. The experiments demonstrate the challenges in using conversational methods with LLMs for even the most prevalent analog circuits.

C. Inability to interact

LLMs' inability to size the transistors and interpret simulation results also makes it impossible to implement analog circuit designs using a conversational approach.

The actual performance of analog circuits is influenced by numerous physical parameters, often requiring the use of complex multi-order models to accurately match simulation results with real-circuit performance. Therefore, simulation results play a critical role in dominating the performance of

GPT-4's response to a 5-T OTA netlist

Input: Please generate a SPICE netlist for a differential input, single-ended output 5-T OTA

Output:

```
* 5-transistor CMOS OTA

.model nmos nmos(level=1 lambda=0 Vto=0.7 KP=120u)
.model pmos pmos(level=1 lambda=0 Vto=-0.7 KP=120u)

M1 out in- 0 N1 nmos W=20u L=2u
M2 N1 in+ 0 0 nmos W=20u L=2u
M3 bias in- Vdd Vdd pmos W =100U l =1U
M4 bias in+ Vdd Vdd pmos W =100U l =1U
M5 bias bias GND GND nmos W =100U l =1U

Vdd vdd 0 DC +5V
GND gnd 0 DC -5V
Vin+ in+ gnd DC .75 AC .25 SIN(0V {sqrt(2)}V {fc})
Vin- in- gnd DC .75 AC -.25 SIN(0V {sqrt(2)}V {fc})
VBias bias GND DC .75

.tran .5us {100/fc}
.ac dec 20 .01Meg fc
.probe ac V(out)
.end
```

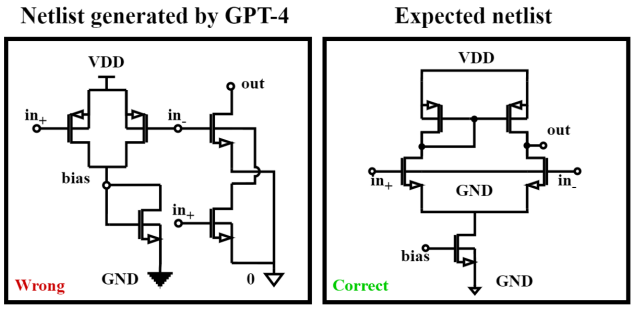


Fig. 2. SPICE netlist file for a 5-transistor OTA generated by GPT4 and comparison between the generated error.

analog circuits. However, the use of a conversational method with LLM does not allow it to read the simulation results and use them for analog circuit design. While it is technically feasible to manually perform tasks such as reading simulation results, sizing transistors, and communicating the results to the LLM, this approach contradicts the objective of attaining full automation of analog circuit auto design.

In conclusion, the conversational design methods with LLMs are currently unable to effectively design analog circuits by automatically reading simulation results and transistor sizing.

D. Decision maker and environment interaction

LLMs' logical ability and understanding of common sense allow them to be better suited for logical reasoning tasks. Many works have investigated applying LLMs as agents to do decision-making tasks.

Yao et al [16] developed ReAct to generate both reasoning traces and task-specific actions. Besides, Qin et al [17] showed equipping LLMs with external tools can expand their capabilities. Karpas et al [18] emphasized the importance of letting LLMs know when and how to utilize external tools which is called an MRKL system. Such approaches of using LLM as a decision-maker and equipping external tools have been used in many domains, such as robotics controller [19] and video games [20].

Similar to the tasks mentioned above, the auto-design problem for analog circuits also involves decision-making (e.g., whether one should decide to retrieve a local knowledge library or to perform a simulation) and interaction with the environment (e.g., obtaining simulation results, transistor sizing) by equipping external tools.

Therefore, in this work, we chose to automate the analog circuit design process by using the LLM agent approach.

III. Overview of LADAC

The flow of LADAC is shown in Fig. 3. LADAC includes user input, agent thinking and action-making, and an action library. A brief illustration of the LADAC's flow is shown below.

The user first inputs the circuit's expected performance metrics and structure description for LADAC's further process. An LLM agent, which is based on GPT4, is applied as a decision-maker to automate the analog circuit design. The actions that the agent can perform include: tuning the devices' parameters and interacting with the simulation result, retrieving the local knowledge library, and further optimizing by a swarm intelligence algorithm, Artificial Bee Colony (ABC) [21]. The local knowledge library consists of manually refined knowledge, which includes basic analog knowledge, multi-stage amplifiers knowledge, and ring oscillator knowledge from papers, online forums, and open-source design materials. The local knowledge library includes text for reference and functions that can be called.

LADAC automates the design of analog circuits by retrieving a local knowledge library to guide the actions of the LLM agent.

A. User input

User input includes a concise circuit description and the expected performance metrics. For example, a user provides a description such as "The parameter W corresponds to the channel width of the input transistor in the first stage." to inform the LLM not only of the circuit's structure but also of the correspondence between the parameters and the transistors. Additionally, the user may specify performance requirements like "I need an amplifier with a gain-bandwidth product (GBW) of 10MHz under a load capacitance of 10pF." to inform the LLM of the expected performance metrics.

LADAC can perform task decomposition based on user input and in conjunction with retrieving the local knowledge library.

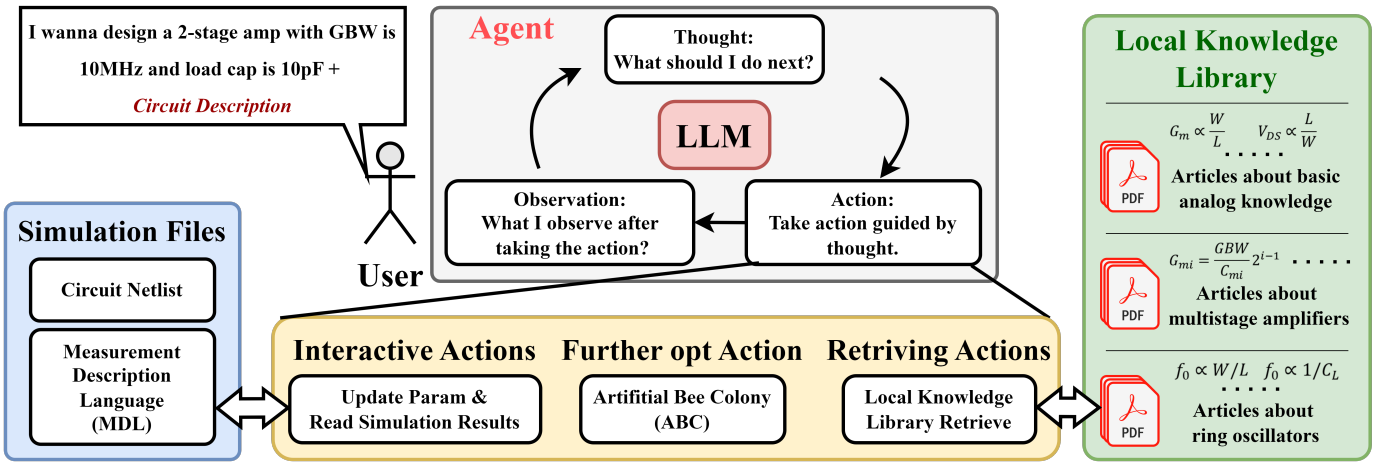


Fig. 3. The flow of LADAC

Examples of a tool and an agent's thought process and actions

```
Tool(
    name = "Update_param",
    func = change_param,
    description = " " "
        Useful for when you think you need to use 'Update_Param' function.
        The input to this tool should be a comma separated list of params'
        name and value to change. E.g., 'W,100n' would be the input
        if you want to change the param 'W' to 100n.
    " " "),
```

Definition of the tool

Description of the conditions for using this tool and input format

Thought: Now I know the width of the input transistor NM0 needs to be 6.502e-06 to meet the target gm. I also need to adjust the width of the tail current transistor NM2 and the PMOS PM0 by the same factor. Then I need to make sure all the transistors of the current stage are in saturation or subthreshold region.
Action: Update_param
Action Input: w1,6.502e-6;w2,6.502e-6;w3,6.502e-6
Observation: Change the param successfully

Agent's thought and action

Fig. 4. An example of a description of an action, and agent's thought, action, and observation

B. LLM agent

We developed an LLM agent, which is based on GPT4 and serves as the central component of LADAC, capable of achieving spontaneous and automated decision-making in analog circuit design. The LLM agent can perform various actions, each with specific indications for their appropriate usage in different situations.

The action indications serve as guidelines for the agent, enabling it to select the most suitable action during the iteration process accurately. Fig. 4 provides a detailed description of a specific action that modifies parameters and the agent's thoughts, action inputs, and observations when performing this action.

The LLM agent follows a systematic approach to address the problem. In the case of designing a two-stage amplifier, LADAC will first retrieve the local knowledge library for the most relevant content and proceed to the next step of the analysis in conjunction with the user's metrics requirements, such as gain-bandwidth product (GBW) and load capacitance (C_L). Subsequently, the agent breaks down the design task into individual stages, focusing on the transconductance (g_m) of each stage. By calculating the expected g_m value for each stage, the agent gains insight into the necessary adjustments required to meet the desired specifications.

To achieve the expected g_m value, the agent performs transistor sizing, optimizing the channel widths and lengths of the transistors in each stage. This sizing procedure is guided by fundamental analog knowledge, incorporating principles and techniques that are related to the transistors, such as the transistor's g_m value is proportional to its channel width-to-length ratio (W/L) [22].

By adopting this step-by-step approach, the LLM agent effectively designs the 2-stage amplifier, ensuring that the transconductance requirements are met through accurate transistor sizing.

C. Actions

As shown in Fig. 3, the knowledge library utilized by LADAC encompasses various categories of actions. These include basic interactive actions, actions for retrieving the local knowledge library to guide top-level design solutions, as well as a further optimizing action which will be illustrated in Part E.

The basic interactive actions enable transistor sizing by modifying the transistor parameters within the netlist file. They facilitate the execution of simulation through the measurement description language (MDL) and subsequently extract the corresponding simulation result file.

The retrieve actions retrieve the content with the highest similarity to the target circuits' description in the local knowledge library and use the results obtained to guide the LLM agent in action planning as well as reasoning.

The actions compensate for the inability of LLMs to interact with circuits and enable the circuit design by retrieving the local knowledge library.

D. Local knowledge library

LADAC's local knowledge library currently includes manually refined knowledge about multi-stage amplifiers, ring oscillators, and basic analog circuits. This information is derived from technical articles, online forums, and open-source design materials. The local knowledge library includes text for reference and functions that can be called. Next, we will briefly describe each part of the local knowledge library.

The basic analog part is derived from the fundamental principles of analog circuits at the transistor level. For instance, the transconductance of a transistor operating in the saturation region is directly proportional to the width-to-length ratio (W/L) of the transistor. This relationship can be quantitatively analyzed and saved as a callable function. Additionally, the drain-source voltage (V_{DS}) of transistors on the same branch is inversely proportional to the W/L ratio. This relationship can be qualitatively analyzed and saved as text that can be retrieved for guidance. By considering these kinds of relationships, the agent can make informed decisions during the sizing process.

Multi-stage amplifier part leverages the existing knowledge of multi-stage amplifiers to guide the design process. It involves utilizing code formulas that can compute the transconductance of each stage based on expected performance. For instance, the transconductance of each stage in a multi-stage amplifier can be defined using the complex-pole method [23] as shown in (1).

$$G_{m,i} = \begin{cases} GBW \cdot C_{m,i} \cdot 2^{(i-1)}, & \text{if not the load stage} \\ GBW \cdot C_L \cdot 2^{(i-1)}, & \text{if the load stage} \end{cases} \tag{1}$$

where $G_{m,i}$ is the transconductance of the ith stage, GBW is the expected gain-bandwidth product, $C_{m,i}$ is the miller capacitor of the ith stage, and C_L is the load capacitor. This relationship can be quantitatively analyzed and saved as a callable function. Furthermore, the multistage amplifier action provides valuable guidance for the agent's subsequent decisions. It prompts the agent to consider the importance of designing multistage amplifiers that necessitate a stage-by-stage transistor sizing approach and ensuring that all transistors within a stage operate in the saturation region before proceeding to design the next stage.

The ring oscillator part is refined from Razavi [12] and some derived knowledge. For example, a ring oscillator's, which is combined with N inverters, oscillation frequency (f_0) is defined by (2).

$$f_0 = \frac{1}{2NT_D} \tag{2}$$

where T_D is the inverters' delay time. Thus, what influences T_D is also concluded in this knowledge part: T_D is proportional to the load capacitor's value and is inversely proportional to the (W/L). This knowledge can effectively guide the agent so that it can tune the devices' parameters according to the current oscillation frequency.

By incorporating this knowledge of analog circuits into the transistor sizing process, the LLM agent can expedite the design process and achieve the desired performance more effectively. This knowledge plays a crucial role in assisting the agent in achieving faster transistor sizing and optimizing the circuit's performance.

E. Further optimization

The primary concern of the LLM is its time-consuming nature during each iteration, which hinders its efficiency in optimizing analog circuits that already satisfy or almost meet the performance requirements.

To solve this problem, we let the agent call a swarm intelligence algorithm, ABC [21]. ABC leverages the circuit parameters previously designed by the LLM as initial values to achieve further circuit optimization through parallel computation. Compared to other optimization problems, optimization problems for circuits tend to make sure they satisfy constrained performance metrics, such as DC gain and GBW, and then optimize other performance metrics, such as circuit area and power. Therefore, the fitness function should be modified to better handle the problem of optimizing constrained metrics and optimization metrics. The modified fitness function is shown in Algorithm 1.

By introducing the ABC algorithm, LADAC can efficiently achieve further circuit optimization at the end of the design flow. Detailed comparisons will be provided in the experimental section.

Algorithm 1 Modified fitness function

Input: Target performance metrics (GBW_{tgt}, PM_{tgt}, $DC_{gain,tgt}$, $Power_{tgt}$), current performance metrics (GBW, PM, DC_{gain}, $Power$)
Output: $Fitness$: A value indicating the circuit's performance.

```
Loss = 0
for Ind in (GBW, PM, DC_gain) do
    if Ind < Ind_tgt then
        Loss+ = max(1e^-6, ((Ind_tgt - Ind)/Ind_tgt)^2)
    else
        Loss+ = 0
    end if
end for
for Ind in (Power) do
    if Ind < Ind_tgt then
        Loss+ = 0
    else
        Loss+ = max(1e^-6, ((Ind_tgt - Ind)/Ind_tgt)^2)
    end if
end for
if Loss==0 then
    Fitness = 10^6
    for Ind in (Power) do
        Fitness* = Ind_tgt/Ind
    end for
else
    Fitness = 5/Loss
end if
```

Multi-stage amp retrieve

Thought:I need to use the Param_guide_multistage_amp tool to get an idea of the gm and miller cap values for each stage.
Action: Param_guide_multistage_amp
Action Input: 10e6, 10e-12, 2
Observation: The target gm of each stage is [0.0001256637061435917 4, 0.001256637061435917 2], the miller cap of each stage is [2e-12]. You must size the transistors stage by stage making sure their gms are within 5 percent error of the expected ones. Before optimizing the next stage, you need to make sure all the transistors of current stage are in saturation region.

Basic analog knowledge retrieve

Thought: The current gm of the input transistors of the first stage is less than the expected gm. I need to use the Gm_tune tool to tune the width to length ratio of the transistor to reach the target gm.
Action: Gm_tune
Action Input: 1e-6,1e-6,1.93259e-5,0.0001256637061435917 4
Observation: the modified width=[‘6.502e-06’] length=[1e-06]…

Fig. 5. LADAC’s representative response

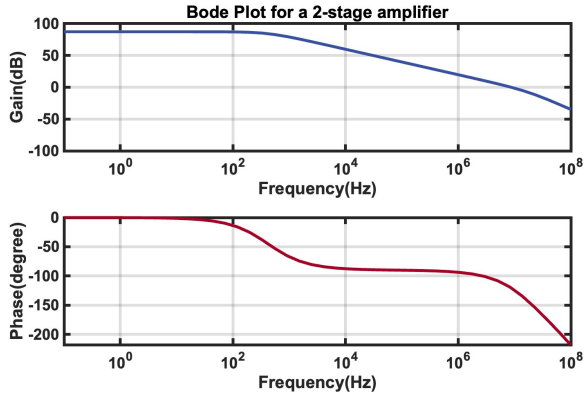


Fig. 6. LADAC-designed 2-stage amplifier’s bode plot

IV. Experiment

In this section, we show the experimental results of LADAC’s automatic design of a 2-stage amplifier, a 3-stage NMCNR amplifier [24], and a ring oscillator [12]. LADAC’s detailed design flows for the 3-stage NMCNR amplifier and the ring oscillator will be separately shown in Part B and Part C. The experimental results show that LADAC can automate the design of analog circuits by reasoning the design flow based on the tools, local knowledge library, and the prompt we provide.

A. Two-stage amplifier

In this part, we employed LADAC in the design of a two-stage amplifier on the $0.18\mu m$ process. Representative LADAC’s response is shown in Fig. 5 which shows how the local knowledge library, including the multi-stage amplifier and basic analog knowledge, assists the agent in reasoning. Guided by the local knowledge library, LADAC achieves designing the 2-stage amplifier by iteratively transistor sizing to meet the expected g_m values.

TABLE I
2-stage amplifier’s performance comparison @$C_L = 10pF$

	Expected value	Performance after LLM	Performance after further optimization
GBW	$\geq 10MHz$	$9.16MHz$	**10.1MHz**
PM	≥ 60	57.5	**60.8**
$Gain_{DC}$	$\geq 80dB$	**81.79dB**	$80.92dB$
I_{dd}	Lower is preferred	$\mathbf{336\mu A}$	$352\mu A$
$\frac{IFOM_S}{\frac{MHz*pF}{mA}}$	Higher is preferred	272.6	**286.9**

To evaluate the performance of the two-stage amplifier, we compared the expected, LADAC initial-designed, and LADAC ultimate-designed, which is further optimized by ABC, performance metrics. These results are depicted in Table I, which demonstrates LADAC’s ability to design and optimize a 2-stage amplifier. To quantitatively evaluate the amplifier’s performance, we utilized I_{dd}-figure of merit for small signal ($IFOM_S$) which is defined in (3). $IFOM_S$ serves as a reliable metric for assessing the performance of an amplifier.

$$IFOM_S = \frac{GBW \cdot C_L}{I_{dd}} \tag{3}$$

According to Table I, the two-stage amplifier after applying the ABC algorithm not only meets the expected performance but also demonstrates a higher $IFOM_S$. This indicates that the additional optimization performed by ABC has not only achieved the desired performance but has also improved the performance of the amplifier. The LADAC-designed 2-stage amplifier’s bode plot is shown in Fig. 6.

B. Three-stage NMCNR amplifier

In this part, we employed LADAC in the design of a three-stage NMCNR amplifier on the $0.18\mu m$ process. The schematic of the amplifier is shown in Fig. 7. Compared to a 2-stage amplifier, a 3-stage NMCNR amplifier adds an extra stage to increase amplification and a nulling resistor to compensate the right-half-plane zero for a more stable structure [24]. Although the 3-stage NMCNR amplifier achieves better performance by introducing additional devices, this also results in it being more difficult to design due to more parameters and loops. Similar to the design of a 2-stage amplifier, LADAC achieves a simplification of a 3-stage NMCNR amplifier design problem by decomposing its design flow by local knowledge library. The design flow is decomposed into poles-zeros separation problems, which further become sizing transistor to the expected gm value problems. The expected, LADAC initial-designed, and LADAC ultimate-designed, 3-stage NMCNR amplifier performance metrics are shown in Table II, which demonstrates LADAC’s ability to design

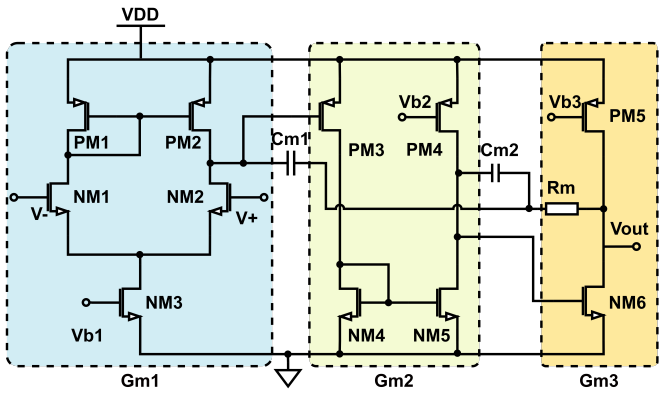


Fig. 7. 3-stage NMCNR amplifier schematic

TABLE II
3-STAGE NMCNR AMPLIFIER'S PERFORMANCE COMPARISON
@$C_L = 10pF$

	Expected value	Performance after LLM	Performance after further optimizaiton
GBW	$\geq 10MHz$	**10.3MHz**	$10.1MHz$
PM	≥ 60	59.8	**60.3**
$Gain_{DC}$	$\geq 120dB$	**131dB**	$126dB$
I_{dd}	Lower is preferred	$592\mu A$	$\mathbf{543\mu A}$
$IFOM_S$ $\frac{MHz*pF}{mA}$	Higher is preferred	168.9	**184.16**

relatively complex analog circuits. The inputs to LADAC and part of the thought process when LADAC designing the 3-stage NMCNR amplifier are shown in the Fig. 8. From Fig. 8, it can be seen that LADAC has the ability of reasoning and decomposing the multi-stage amplifier design problem by local knowledge library. The LADAC-designed 3-stage NMCNR amplifier's bode plot is shown in Fig. 9.

LADAC's successful design of 2-stage amplifiers as well as 3-stage NMCNR amplifiers confirms its ability to design analog circuits of the amplifier type.

C. Ring oscillator

To demonstrate LADAC's feasibility to design other analog circuits, we applied LADAC to design a ring oscillator combined by three inverters [12]. Its schematic is shown in Fig. 10. The inverter's time delay (T_D) is inversely proportional to its width to length ratio (W/L) and is proportional to its load cap. The ring oscillator's, which is combined by N inverters [12], oscillation frequency (f_0) is shown in (2). Thus, in this experiment, we informed LADAC that the f_0 is proportional to the transistor width-to-length ratio and inversely proportional to the load capacitance.

We requested LADAC to design a ring oscillator with f_0 of 100 MHz as the ultimate goal. LADAC approximates the

LADAC's process for designing a 3-stage NMCNR amp

Input
I wanna design a 3-stage NMCNR amp with GBW is 10MHz and load cap is 10pF, can you tell me the value of each stage's gm ,miller cap and nulling res?
The basic flow is 1: 'Pram_guide_multistage_amp' to get gm of each stage
2: size the transistors stage by stage.
After sizing the transistors to the expected gm, you need to make sure each transistor working in region 2 and check their gm is really close to the target
+Description of the circuit structure

Param_Guide_Multistage_Amp
Action Input: 10e6,10e-12,3
Observation: the target gm of each stage is [0.00012566370614359174, 0.0002513274122871835, 0.0025132741228718345], the miller cap of each stage is [2e-12, 2e-12] if the amplifier's type is NMCNR the value of the nulling resistor is 397.88735772973837
Thought: I need to read the current transconductance (gm) of the input transistor NM0 to compare it with the target gm value for the first stage.

Read_DC_oppoint
Observation: {'gm_NM0_I0': '1.93151e-05'}
Thought: The current transconductance (gm) of NM0 is 1.93151e-05, which is not close to the target gm value for the first stage (0.00012566370614359174). I need to adjust the width of NM0 to achieve the target gm. I will use the Gm_tune tool to calculate the new width for NM0.

Gm_Tune
Action Input: 1e-6,1e-6,1.93151e-05,0.00012566370614359174
Observation: the modified width=['6.506e-06'] length=[1e-06]
Thought: The Gm_tune tool suggests that the width of NM0 should be adjusted to approximately 6.506e-06 meters (6.506 µm) to achieve the target transconductance, while keeping the length at 1e-06 meters (1 µm). I also need to adjust the width of the load and tail current transistors by a factor of 6.505982684200016 times their original width to maintain proper operation.

Update_param
Action input: w1,6.506e-6;w2,6.506e-6;w3,6.506e-6
Observation: Change the param successfully

⋮

End
The gm values for each stage after tuning are approximately 0.000119906 for the first stage, 0.000294061 for the second stage, and 0.0028287 for the third stage. The miller capacitance values are 2e-12 for both the first and second stages. The nulling resistor value is 397.88735772973837 ohms. The widths of the transistors have been adjusted to approximately 6.506 µm for the first stage, 17.18 µm for the second stage, and 89.53 µm for the third stage. The lengths of the transistors remain at 1 µm for all stages.

Fig. 8. LADAC designing 3-stage NMCNR amplifier

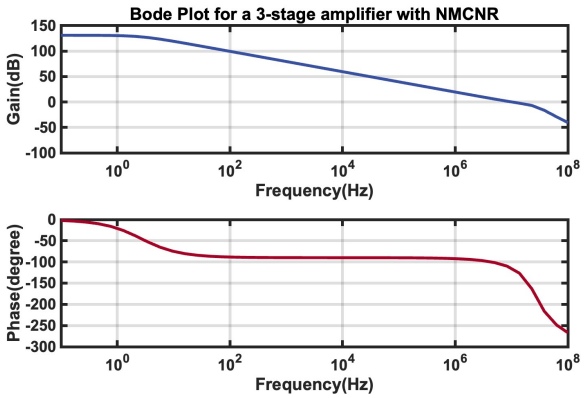


Fig. 9. LADAC-designed 3-stage NMCNR amplifier's bode plot

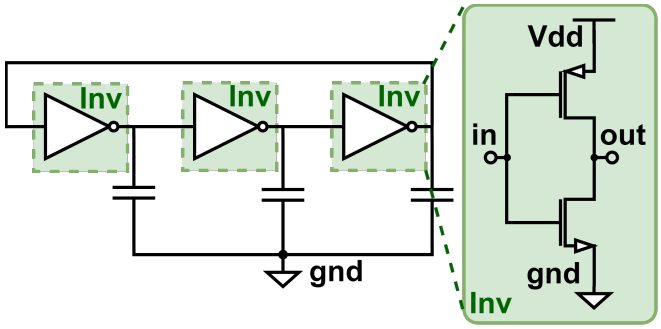


Fig. 10. Ring oscillator schematic

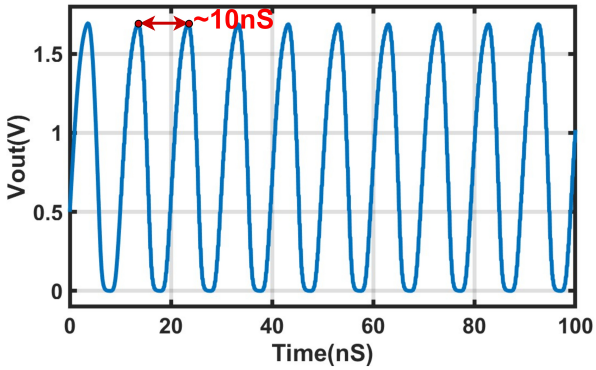


Fig. 11. Ring oscillator output signal waveform

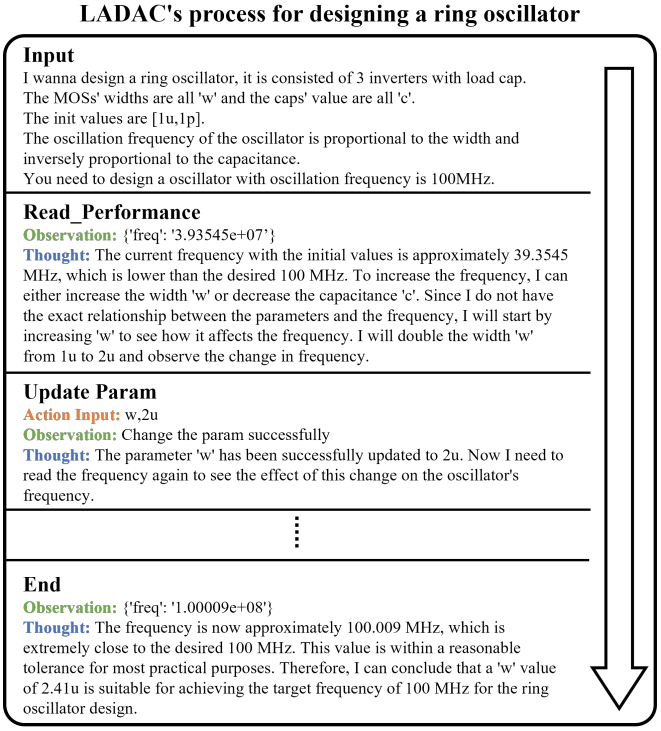


Fig. 12. LADAC designing the ring oscillator

target f_0 by continuously sizing the transistors and tuning the load caps. The LADAC-designed ring oscillator's f_0 is $100.009MHz$ which is very close to the expected value and LADAC thinks it does not need ABC for further optimization. The LADAC-designed ring oscillator's output signal waveform is shown in Fig. 11 and the LADAC's response is shown in Fig. 12.

From the above experiments, we can see that LADAC has successfully implemented the automatic design of two kinds of amplifiers as well as a ring oscillator by retrieving the local knowledge library and interacting with the simulation tool. This also demonstrates the promise of the LLMs for the design of more complex analog circuits.

V. Conclusion

In this study, we have presented LADAC, which is a new approach utilizing an LLM agent equipped with basic analog knowledge library and simulation interfaces to achieve the auto design of analog circuits. We specifically showcase its effectiveness in auto design of a two-stage amplifier, a 3-stage NMCNR amplifier, and a ring oscillator. Our work demonstrate the promising potential of LLMs in the field of analog circuit design. The LLMs faces the following challenges and opportunities in analog circuit design.

Challenges: The local analog knowledge library for this work still needs to be refined manually, and the use of materials such as technical articles directly as a local knowledge library for LLM to retrieve often does not refine the most accurate information for the design and thus does not allow for an automated design process. To realize a wider range of circuit designs and avoiding human involvement, there should be a methodology that extracts an accurate knowledge library from the raw data, such as technical papers, online forums, and open source design materials to guide the LLM agent towards a subsequent auto design process. The methodology should also support self-calibration to confirm its correctness.

In addition, the possibility of automating the design of more complex analog circuits, especially system-level analog circuits such as analog to digital converters (ADCs) and compute in memory (CIM) circuits, through one LLM agent still needs to be experimentally verified. New LLM agent structures, such as multiple agents working together, may then be needed to realize designs for more complex circuits.

Opportunities: Based on the LLMs' ability to understand natural language, we can summarize the application scenarios of each circuit in the local knowledge library. This can enable the LLMs to achieve the selection of the most appropriate circuit structure according to the user's needs. As a result, the LLM holds the promise of not only enabling the selection of basic circuit modules, such as amplifiers and comparators, for a given application scenario, but also of structurally optimizing each module in the system-level circuitry to achieve more outstanding performance.

Moving forward, our future work aims to expand upon this research by developing a comprehensive analog knowledge library to guide LLM agents and achieve a wider range of analog circuits. This will involve leveraging the capabilities of the LLM agents to automatically generate and refine the analog knowledge from various ways, such as papers, online forums, and open-source design materials. This work holds

the potential to further advance the automation and efficiency of the LLM auto-design analog circuits process.

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